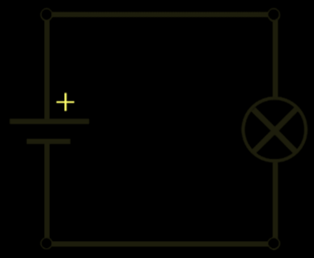
Electrical Voltage

by ZAKARIA BENHAIRA

Considering the following circuit where a generator connected in series with a lamp :



The generator provides a quantity of electrical energy to the lamp which is equal to the diffrence between its poles :

$$E_e = E_+ - E_-$$
$$E_e = q(V_+ - V_-)$$

This formula has an issue it is charge dependent, which means to use we have to calculate the charge : $q = It$.

To avoid this we instead divide by the charge to get a quantity called voltage :

$$\frac{E_e}{q} = \frac{q(V_+ - V_-)}{q} = V_+ - V_- \stackrel{\text{def}}{=} U$$

Voltage (U) is the energy difference delivered by the generator per charge : $U = \dfrac{\Delta E}{q}$.

And because voltage is a diffrence between to points which a are the poles of the generator to measure it we do so by a device connected in parallel.